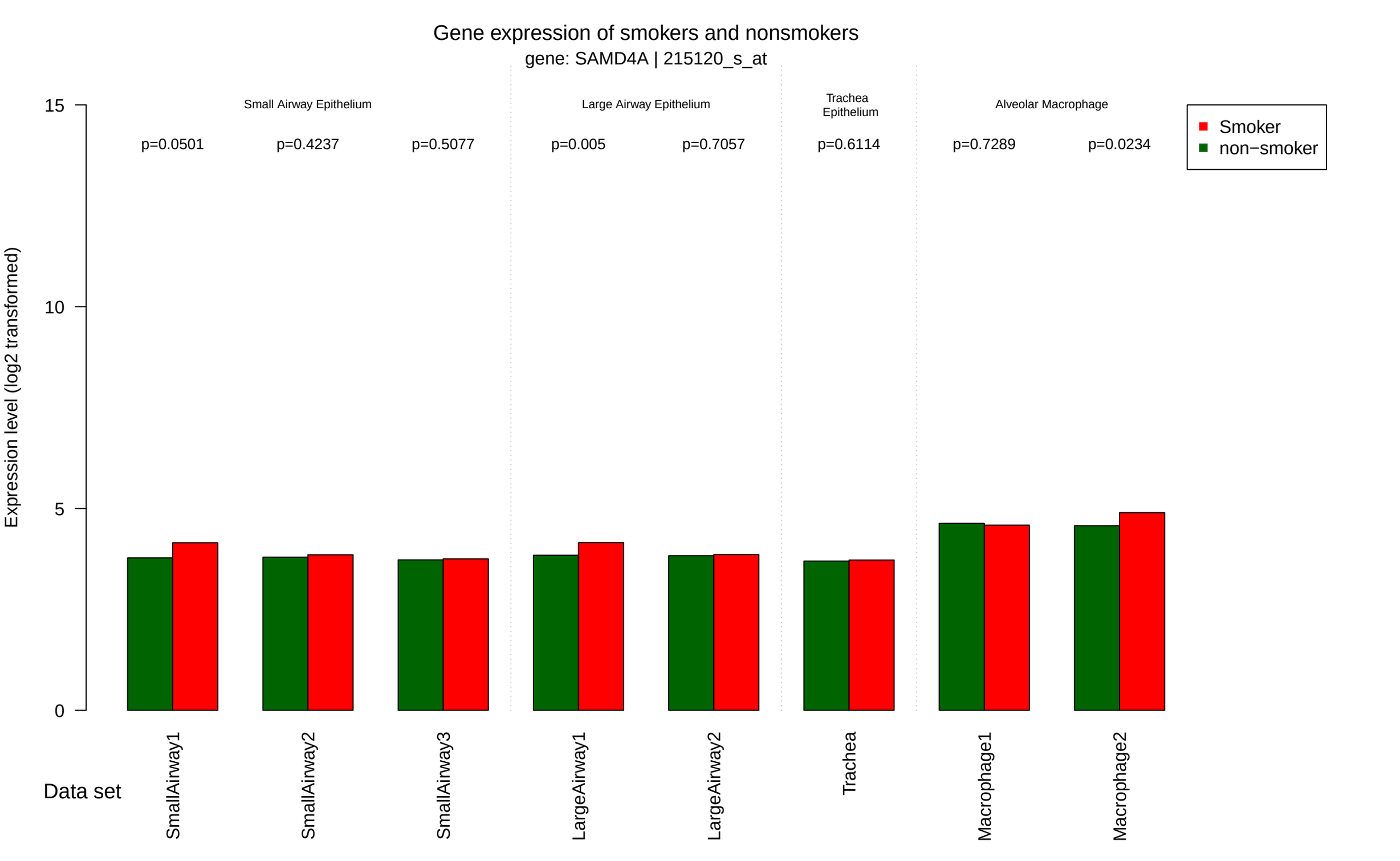
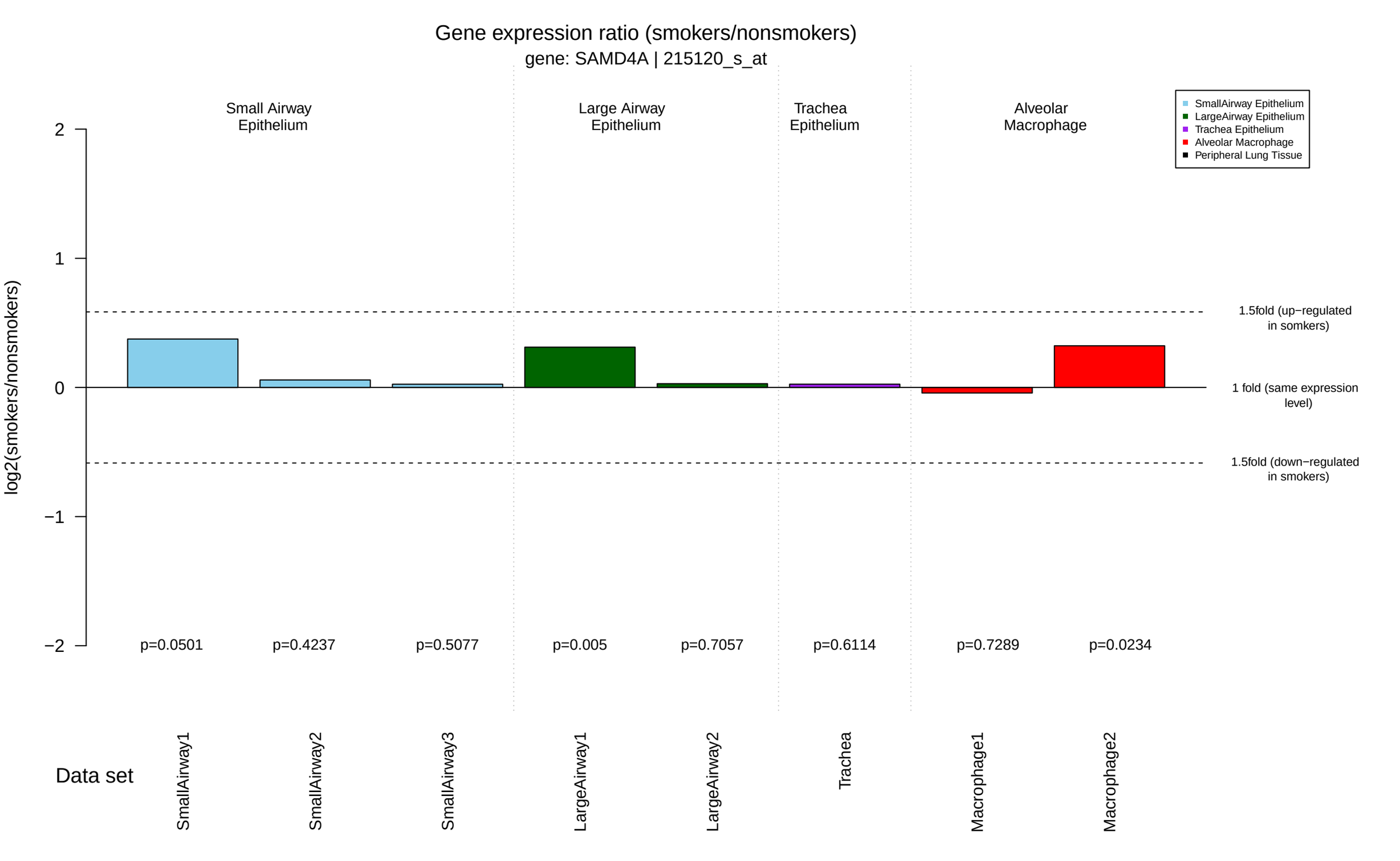
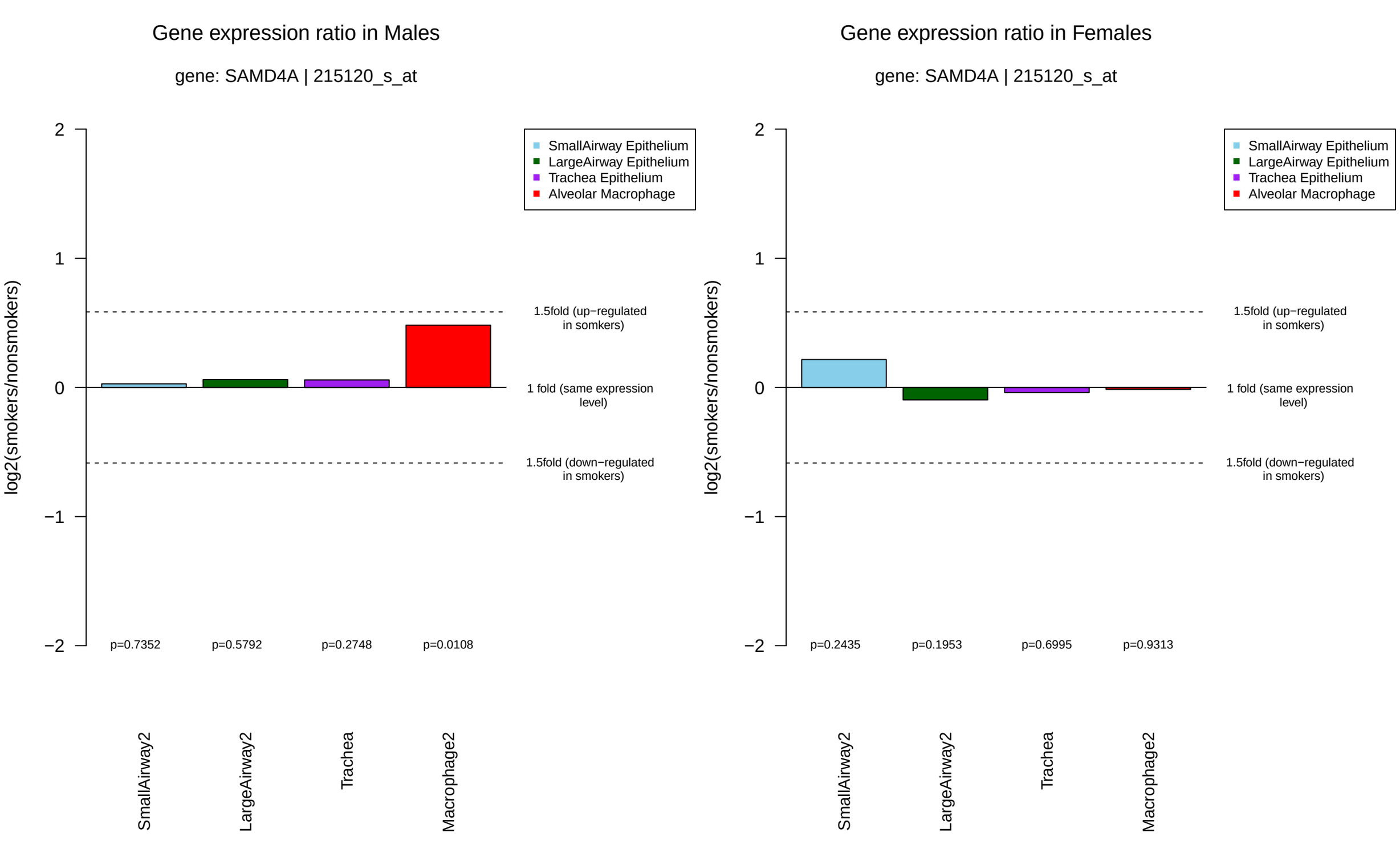
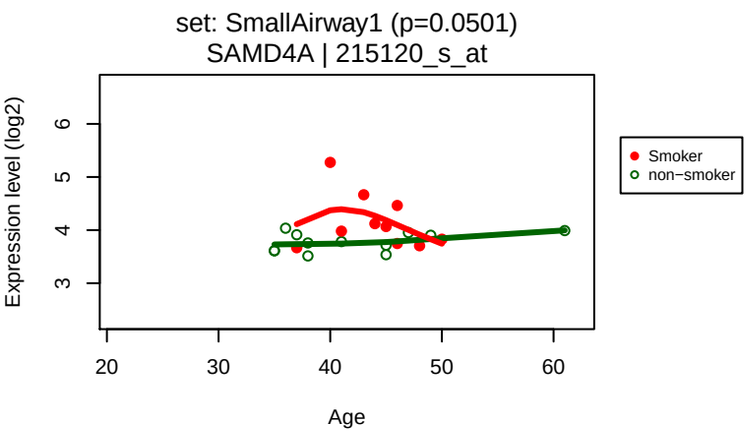
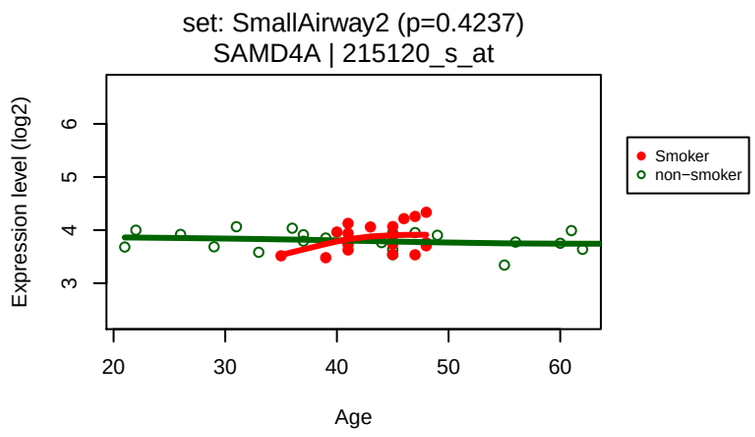
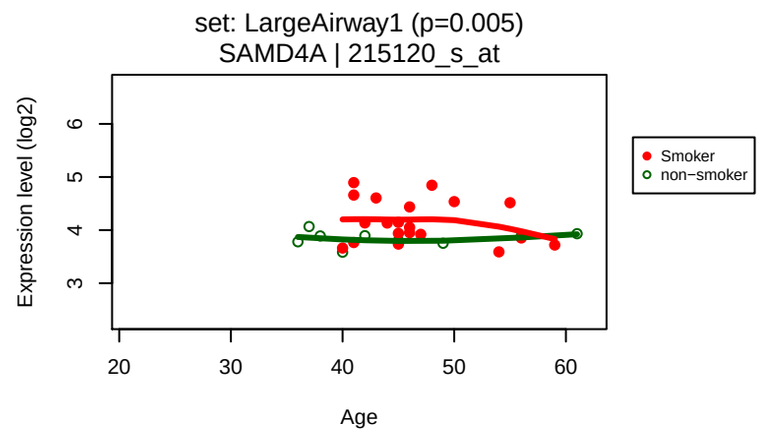

Gene expression of smokers and nonsmokers
gene: SAMD4A | 215120_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0501
p=0.4237
p=0.5077
p=0.005
p=0.7057
p=0.6114
p=0.7289
p=0.0234
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

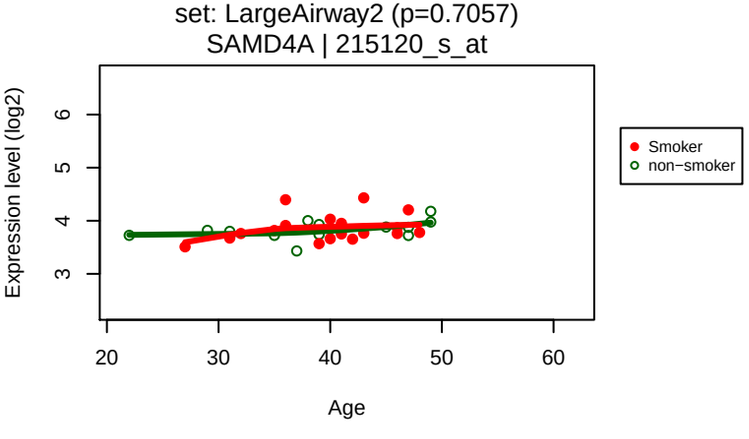

Gene expression ratio (smokers/nonsmokers)
gene: SAMD4A | 215120_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0501
p=0.4237
p=0.5077
p=0.005
p=0.7057
p=0.6114
p=0.7289
p=0.0234
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

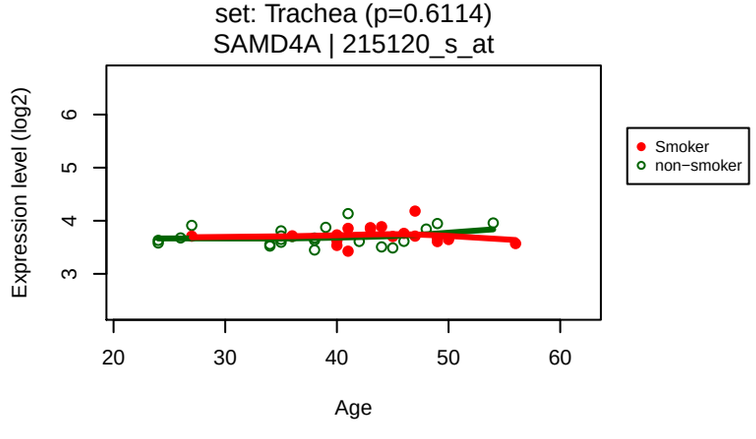

Gene expression ratio in Males
gene: SAMD4A | 215120_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7352
p=0.5792
p=0.2748
p=0.0108
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SAMD4A | 215120_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2435
p=0.1953
p=0.6995
p=0.9313
SmallAirway2
LargeAirway2
Trachea
Macrophage2

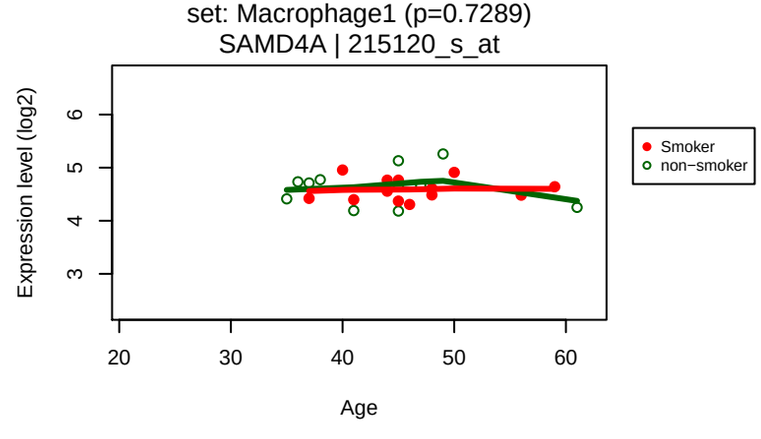

set: SmallAirway1 (p=0.0501)
SAMD4A | 215120_s_at
Expression level (log2)
Age
Smoker
non−smoker

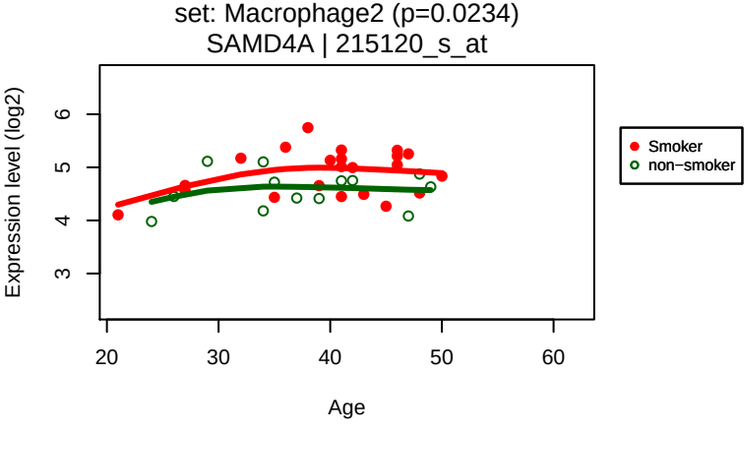

set: SmallAirway2 (p=0.4237)
SAMD4A | 215120_s_at
Expression level (log2)
Age
Smoker
non−smoker

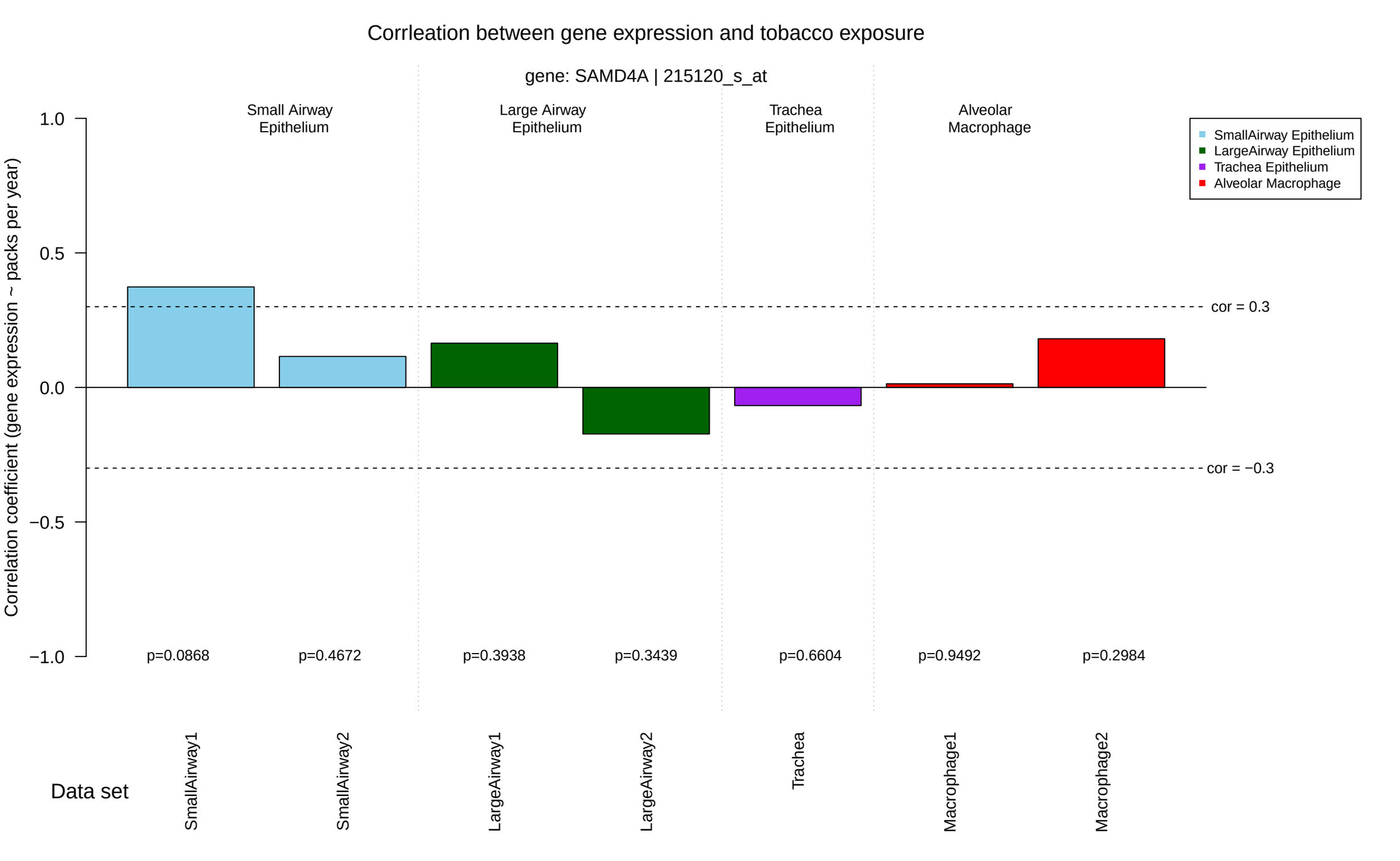

set: LargeAirway1 (p=0.005)
SAMD4A | 215120_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7057)
SAMD4A | 215120_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6114)
SAMD4A | 215120_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7289)
SAMD4A | 215120_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0234)
SAMD4A | 215120_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SAMD4A | 215120_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0868
p=0.4672
p=0.3938
p=0.3439
p=0.6604
p=0.9492
p=0.2984
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2